Quiz 2 - DSC 10, Fall 2025

Full Name:

PID:

Quiz Time: ◯ 2PM ◯ 3PM ◯ 4PM ◯ 5PM

Instructions:

- This quiz consists of 5 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.
- Assume we have already run `import babypandas as bpd` and `import numpy as np`.

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

The DataFrame `chipotle` contains information about all ingredients on the menu at the Chipotle restaurant chain. The DataFrame is indexed by ingredient `"name"` (str), and also includes each ingredient's `"calories"` (int), `"protein"` content in grams (str), `"type"` (str: either `"meat"`, `"base"`, or `"veggie"`), and `"spice_level"` (int: either 0, 1, 2, or 3). The `chipotle` DataFrame has 30 rows, with the first 5 shown.

name	calories	protein	type	spice_level
guacamole	195	3 g	veggie	0
barbacoa	180	32 g	meat	2
steak	150	35 g	meat	3
black beans	130	9 g	base	0
white rice	190	4 g	base	0

Question 1

For a typical diet, the recommended daily protein intake is 80 g. Fill in the blanks in the code below to add a new column to `chipotle` that contains values 0 to 100 representing the percentage of the recommended daily protein intake each ingredient provides.

```
def clean_protein(grams):
    return int(grams.split(" ")[__(a)__]) / __(b)__ * 100
chipotle = chipotle.assign(protein_percent = __(c)__)
```

(a):

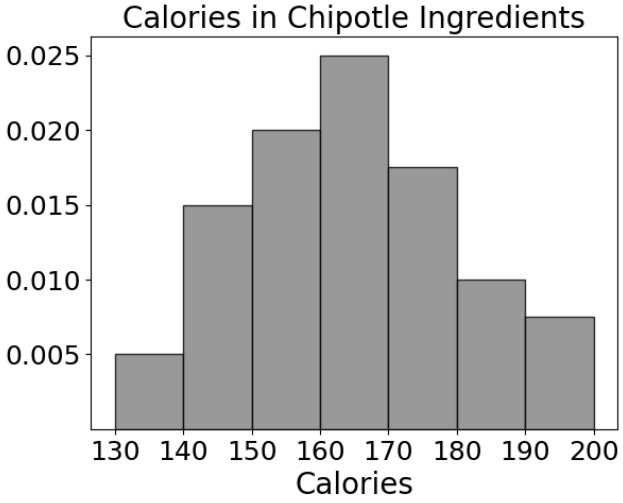
(b):

(c):

Question 2

Below is a density histogram showing the distribution of calories for all 30 ingredients in `chipotle`.

a) How many ingredients have less than 160 calories?



b) Suppose we had plotted the histogram with the argument `bins=[120, 160, 190, 210]`.

How many bars would the histogram have?

What would the height of the leftmost bar be?

PID: _____

Question 3

What is the maximum possible number of rows in the DataFrame that results from the expression below?

```
chipotle.groupby(["spice_level", "type"]).mean()
```

Question 4

Our tutor Noah ate at Chipotle ten times last month (he loves Chipotle). Each time, Noah only ordered one item, either guacamole, barbacoa, or steak. The DataFrame on the right, called `noahs_meals`, shows what he ordered. Suppose we run the code below.

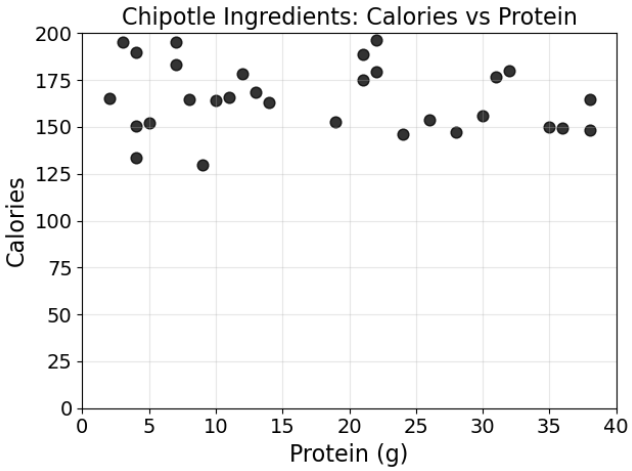
```
first_five = chipotle.take(np.arange(5))
first_five.merge(noahs_meals, on = "type")
```

How many rows does the resulting DataFrame contain?

	item	type
0	guacamole	veggie
1	steak	meat
2	guacamole	veggie
3	barbacoa	meat
4	barbacoa	meat
5	guacamole	veggie
6	barbacoa	meat
7	steak	meat
8	steak	meat
9	barbacoa	meat

Question 5

a) Use the scatterplot of all 30 ingredients to determine how many have a calorie-to-protein ratio that is at least 5.



b) Let `calories` and `protein` be `numpy` arrays containing the calorie and protein amounts, as ints, for each of the 30 ingredients in `chipotle`. Fill in the blanks so that the code below calculates the number of ingredients with a calorie-to-protein ratio of at least 5.

```
count = __(a)__
for i in np.arange(__(b)__):
    if calories[i] / protein[i] >= 5:
        count = __(c)__
```

(a): (b): (c):

Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.